ADITYA J. VALAKI

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Summary

Final-year B.Tech IT student (CGPA 8.20, graduating 2027) in **data science, software developer, ML, and generative AI**. Led an 11-engine LLM/OCR benchmark at DA-IICT (**86.1% F1**), built deep learning and NLP classifiers and an F1 telemetry pipeline, and deployed models with FastAPI and Streamlit.

Technical Skills

Languages: Python, SQL, JavaScript, Java, C++, C#, HTML/CSS
Stats & Simulation: hypothesis testing (Wilcoxon), Monte Carlo simulation, backtesting, cross-validation, probabilistic forecasting (Brier score)
ML / DL: scikit-learn, TensorFlow/Keras, CNNs, transfer learning (EfficientNetB0), Random Forest, ensembles, regression, classification, feature engineering, model evaluation (precision, recall, F1)
GenAI / NLP: LLM APIs (Gemini, GPT-4o, Claude), multimodal LLM benchmarking, prompt engineering, LangChain, LangGraph, LangSmith Fleet, Deep agents, OCR
Data & BI: pandas, NumPy, SciPy, DuckDB, Parquet, EDA, data cleaning, ETL pipelines, time-series data, RapidFuzz, Power BI, IBM Cognos, Excel, matplotlib, seaborn
Software: FastAPI (REST APIs), React, Next.js, React Native, Vite, Supabase, Firebase, Git, pytest, pyInstaller, and model deployment
Learning: Docker, RAG with vector databases

Experience

Summer Research Intern, Applied AI | *DA-IICT, Gandhinagar (Prof. Dr. Anupam Rana)* May 2026 – Jul 2026
- Designed a controlled benchmark of **11 AI reading engines** (7 cloud multimodal LLMs, 3 on-device VLMs, Tesseract baseline) for handwritten Indian prescriptions: 50 pharmacist-verified prescriptions, 192 medicines.
- Built the Python evaluation pipeline (RapidFuzz matching, per-item precision, recall, F1) and validated the ranking with Wilcoxon signed-rank tests.
- Found Gemini 3.5 Flash best (**86.1% F1**, 90.1% recall) and on-device models not viable (<24% F1), which set the product architecture; failure analysis made human verification and confidence gating part of the design.

Business Analyst Intern | *Cognifyz Technologies, Nagpur* May 2025 – Jul 2025
- Ran EDA and correlation analysis (pandas, seaborn) on a financial investment dataset; built dashboards and presented recommendations to non-technical stakeholders.

Business Development Executive Intern | *Zeeniith Technology (on-site)* Dec 2025 – Aug 2026
- Worked on the commercial side end to end: lead sourcing, client outreach, pricing, proposals, and follow-through until delivery. Tracked pipeline and outreach data to decide where sales effort went, and reported conversion and delivery status back to the team.
- Built and segmented the prospect list of engineering colleges for ScholarPlace, an AI-integrated B2B education platform, to target outreach.

Projects

Formula 1 Race Strategy & Telemetry Analytics | *Python, FastF1, DuckDB, FastAPI, React* Aug 2026 – Present
- Built an end-to-end Windows and browser app replaying every F1 session since 2023, from FastF1 ingest and a Parquet/DuckDB data lake to a FastAPI backend and React interface, with 900+ automated tests.
- Modelled tyre wear and fuel burn with a linear lap-time model, feeding a lap-by-lap race simulator that ranks pit strategies by finishing position.
- Backtested Monte Carlo race predictions on 87 races, beating the starting grid on rank correlation (0.69 vs 0.66) and, over 2023–25, on win-probability Brier score (0.54 vs 0.63; Wilcoxon $p < 0.001$).

Medication AI – Prescription Reader & Reminder App | *FastAPI, LLM APIs, React Native* May 2026 – Aug 2026
- Served the winning multimodal LLM through a FastAPI backend with a fuzzy-matching correction layer and confidence-based verification routing, Co-developed the React Native app: prescription capture, per-medicine confirmation, dosage-aware reminders and one-tap SOS.

Visual Malware Detector v3 | *TensorFlow/Keras, scikit-learn, Streamlit, Joblib* Dec 2025 – Apr 2026
- Built a three-layer hybrid detector: an Authenticode-aware Shannon-entropy pre-filter, an **EfficientNetB0 CNN** (transfer learning) on grayscale byte-plot images (Malimg, 25 malware families) and a **Random Forest** on 69 PE-header features (ClaMP).
- Combined both models through weighted score fusion for more robust family prediction than either model alone.
- Paper submitted to NIIS 2026, International Conference on Next-Gen Integrated and Intelligent Systems.

FakeShield – Fake Content Detection | *Python, NLP, scikit-learn* Dec 2025 – Mar 2026
- Built an NLP pipeline for misinformation detection: text cleaning, tokenisation, stop-word removal and TF-IDF vectorisation feeding supervised classifiers.

Education

B.Tech, Information Technology | *G H Patel College of Engg. & Tech. (GCET), CVM Univ.* Jun 2023–Jun 2027
CGPA **8.20/10**, no backlogs, NBA-accredited program Vallabh Vidyanagar

Higher Secondary (12th Science, Mathematics) | *Shree Gyanmanjari Vidyapith* – 74.75 PR 2021–2023, Bhavnagar

Certifications & Activities

- IBM Data Analyst Professional Certificate (Coursera); LangChain Academy (LangChain, LangGraph, LangSmith Fleet, Deep Agents); NPTEL: Cloud, IoT.
- IEEE GCET SB Joint Treasurer and Branch Coordinator; IEEE R10 ACEI Entrepreneurship Ambassador; ISTE Sub-Core Member.